Ch. 25 HW # 2 Additional Problems

Principles of Macroeconomics

January 30, 2026

1. Eastland College is concerned about the rising price of textbooks that students must purchase. To better identify the increase in the price of textbooks, the dean asks you, the Economics Department's star student, to create an index of textbook prices. The average student purchases three English, two math, and four economics textbooks. The prices of these books are given in the accompanying table.

	2002	2003	2004
English textbook	\$50	\$55	\$57
Math textbook	70	72	74
Economics textbook	80	90	100

   (a) Assume the typical student buys one of each textbook. Create the price index for these books for all years with a base year of 2002.

   (b) What is the percent change in the price of an English textbook from 2002 to 2004?

   (c) What is the percent change in the price of a math textbook from 2002 to 2004?

   (d) What is the percent change in the price of an economics textbook from 2002 to 2004?

   (e) What is the percent change in the market index from 2002 to 2004?

2. Fill in the table below. Does the family get worse off or better off from 2020-2025?

Year	Nominal Family Income	CPI	Real Family Income
2020	\$40,000	100	
2025	\$44,000	125	

3. A report found that the real entry-level wage for college graduates declined by 5% between 2015 and 2022. The nominal entry level wage in 2022 was \$14.35 per hour. The CPI was 237.0 in 2015 and 292.7 in 2022.

(a) What was the real entry-level wage in 2022?

(b) What was the real entry-level wage in 2015?

(c) What was the nominal entry-level wage in 2015?

(d) What was the annual inflation rate from 2015 to 2022?